

# 20261013 Quiz

for (MATH301) Modern Algebra I

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**Exercise 1.7.18.** Let  $H$  be a group acting on a set  $A$ . Prove that the relation  $\sim$  on  $A$  defined by

$$a \sim b \iff a = h \cdot b \text{ for some } h \in H$$

is an equivalence relation. For each  $x \in A$ , the equivalence class of  $x$  under  $\sim$  is called the orbit of  $x$  under the action of  $H$ . The orbits under the action of  $H$  partition the set  $A$ .

*Solution.* For every  $a \in A$ ,

$$a = 1 \cdot a,$$

so  $a \sim a$ . If  $a \sim b$ , then  $a = h \cdot b$  for some  $h \in H$ , and hence

$$b = h^{-1} \cdot a,$$

so  $b \sim a$ . Finally, if  $a \sim b$  and  $b \sim c$ , say  $a = h \cdot b$  and  $b = k \cdot c$ , then

$$a = h \cdot (k \cdot c) = (hk) \cdot c,$$

so  $a \sim c$ . Thus  $\sim$  is an equivalence relation.

For  $x \in A$ , its equivalence class is

$$[x] = \{h \cdot x \mid h \in H\} = H \cdot x,$$

which is precisely the orbit of  $x$ . Hence the orbits partition  $A$ .

□

**Exercise 2.2.4.** For each of  $S_3$ ,  $D_4$ , and  $Q_8$ , compute the centralizers of each element and find the center of each group. Does Lagrange's Theorem (Exercise 1.7.19) simplify your work?

*Notation.* Following MATH000,  $D_n$  denotes the dihedral group of symmetries of a regular  $n$ -gon, so  $|D_n| = 2n$ . Thus  $D_4$  here is denoted by  $D_8$  in Dummit–Foote.

*Solution.* In  $S_3$ , the identity commutes with every element. A transposition commutes only with itself and the identity, while the two 3-cycles commute with each other. Thus

| $x$                | $C_{S_3}(x)$            |
|--------------------|-------------------------|
| 1                  | $S_3$                   |
| $(12), (13), (23)$ | $\langle x \rangle$     |
| $(123), (132)$     | $\langle (123) \rangle$ |

and therefore

$$Z(S_3) = \{1\}.$$

Write

$$D_4 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle.$$

Then  $r^2$  is central, rotations commute with one another, and the reflections split into two commuting pairs. Hence

| $x$        | $C_{D_4}(x)$              |
|------------|---------------------------|
| $1, r^2$   | $D_4$                     |
| $r, r^3$   | $\langle r \rangle$       |
| $s, sr^2$  | $\langle s, r^2 \rangle$  |
| $sr, sr^3$ | $\langle sr, r^2 \rangle$ |

so

$$Z(D_4) = \{1, r^2\}.$$

In  $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ , the elements  $\pm 1$  are central, and each of  $\pm i$ ,  $\pm j$ , and  $\pm k$  commutes precisely with the corresponding cyclic subgroup of order 4. Thus

| $x$     | $C_{Q_8}(x)$        |
|---------|---------------------|
| $\pm 1$ | $Q_8$               |
| $\pm i$ | $\langle i \rangle$ |
| $\pm j$ | $\langle j \rangle$ |
| $\pm k$ | $\langle k \rangle$ |

and

$$Z(Q_8) = \{\pm 1\}.$$

Lagrange's Theorem simplifies these computations because  $C_G(x) \leq G$  and  $\langle x \rangle \leq C_G(x)$ , so the possible orders of a centralizer are strongly restricted once one finds an element that does not commute with  $x$ .

□

**Exercise 2.2.6.** Let  $H$  be a subgroup of the group  $G$ .

- (a) Show that  $H \leq N_G(H)$ . Give an example to show that this is not necessarily true if  $H$  is not a subgroup.
- (b) Show that  $H \leq C_G(H)$  if and only if  $H$  is abelian.

*Solution.*

- (a) Let  $h \in H$ . Since  $H$  is a subgroup,  $h x h^{-1} \in H$  for every  $x \in H$ , so

$$h H h^{-1} \subseteq H.$$

Applying the same argument to  $h^{-1}$  gives the reverse inclusion. Hence  $h H h^{-1} = H$ , so  $h \in N_G(H)$ . Therefore

$$H \leq N_G(H).$$

The conclusion can fail when  $H$  is only a subset. In  $S_3$ , let

$$H = \{(12), (13)\}.$$

Then  $(12) \in H$ , but

$$(12)H(12)^{-1} = \{(12), (23)\} \neq H.$$

Thus  $(12) \notin N_{S_3}(H)$ , so  $H \not\leq N_{S_3}(H)$ .

- (b) If  $H \leq C_G(H)$ , then every element of  $H$  commutes with every element of  $H$ , so  $H$  is abelian. Conversely, if  $H$  is abelian, then every  $h \in H$  commutes with every element of  $H$ , hence  $h \in C_G(H)$ . Therefore

$$H \leq C_G(H) \iff H \text{ is abelian.}$$

□

**Exercise 2.4.3.** Prove that if  $H$  is an abelian subgroup of a group  $G$ , then  $\langle H, Z(G) \rangle$  is abelian. Give an explicit example of an abelian subgroup  $H$  of a group  $G$  such that  $\langle H, C_G(H) \rangle$  is not abelian.

*Solution.* Every element of  $Z(G)$  commutes with every element of  $G$ , and the elements of  $H$  commute with one another. Hence every word in elements of  $H \cup Z(G)$  can be rearranged into the form  $hz$  with  $h \in H$  and  $z \in Z(G)$ . For  $h_1, h_2 \in H$  and  $z_1, z_2 \in Z(G)$ ,

$$(h_1 z_1)(h_2 z_2) = h_1 h_2 z_1 z_2 = h_2 h_1 z_2 z_1 = (h_2 z_2)(h_1 z_1).$$

Thus  $\langle H, Z(G) \rangle$  is abelian.

For a counterexample, let  $G = D_4$  and  $H = \langle r^2 \rangle = Z(D_4)$ . Then  $H$  is abelian and

$$C_{D_4}(H) = D_4.$$

Therefore

$$\langle H, C_{D_4}(H) \rangle = D_4,$$

which is not abelian.

□

**Exercise 2.4.7.** Prove that the subgroup of  $S_4$  generated by (12) and (13)(24) is isomorphic to  $D_4$ , the dihedral group of order 8.

*Solution.* Let

$$a = (12), \quad b = (13)(24), \quad r = ab, \quad s = a.$$

Both  $a$  and  $b$  have order 2, and  $r = ab$  is a 4-cycle, so  $|r| = 4$ . Also

$$s^2 = 1$$

and

$$srs = a(ab)a = ba = (ab)^{-1} = r^{-1}.$$

Thus  $r$  and  $s$  satisfy the defining relations of  $D_4$ .

Moreover,

$$\langle a, b \rangle = \langle r, s \rangle,$$

since  $s = a$  and  $b = sr$ . The relations above reduce every element of  $\langle r, s \rangle$  to one of

$$1, r, r^2, r^3, s, sr, sr^2, sr^3.$$

Hence  $|\langle r, s \rangle| \leq 8$ . Since  $\langle r \rangle$  has order 4 and the transposition  $s$  is not in  $\langle r \rangle$ , the cosets  $\langle r \rangle$  and  $s\langle r \rangle$  are distinct, so  $|\langle r, s \rangle| \geq 8$ . Therefore  $|\langle r, s \rangle| = 8$ . The assignment from the usual generators of  $D_4$  to  $r$  and  $s$  defines a surjective homomorphism

$$D_4 \rightarrow \langle r, s \rangle.$$

Since both groups have order 8, this homomorphism is an isomorphism.

□

**Exercise 2.4.14.** A group  $H$  is called finitely generated if there is a finite set  $A$  such that  $H = \langle A \rangle$ .

- (a) Prove that every finite group is finitely generated.
- (b) Prove that  $\mathbb{Z}$  is finitely generated.
- (c) Prove that every finitely generated subgroup of the additive group  $\mathbb{Q}$  is cyclic.
- (d) Prove that  $\mathbb{Q}$  is not finitely generated.

*Hint.* For part (c), if  $H$  is a finitely generated subgroup of  $\mathbb{Q}$ , show that  $H \leq \langle 1/k \rangle$ , where  $k$  is the product of all denominators appearing in a set of generators for  $H$ .

*Solution.*

- (a) If  $H$  is finite, then  $H = \langle H \rangle$ , and  $H$  itself is a finite generating set. Hence every finite group is finitely generated.
- (b) Under addition,

$$\mathbb{Z} = \langle 1 \rangle,$$

so  $\mathbb{Z}$  is finitely generated.

- (c) Suppose

$$H = \left\langle \frac{a_1}{b_1}, \dots, \frac{a_n}{b_n} \right\rangle \leq \mathbb{Q},$$

where each  $b_i > 0$ , and let

$$k = b_1 b_2 \cdots b_n.$$

For each  $i$ ,

$$\frac{a_i}{b_i} = a_i \frac{k}{b_i} \cdot \frac{1}{k} \in \left\langle \frac{1}{k} \right\rangle.$$

Hence

$$H \leq \left\langle \frac{1}{k} \right\rangle.$$

Since every subgroup of a cyclic group is cyclic,  $H$  is cyclic.

- (d) If  $\mathbb{Q}$  were finitely generated, then part (c) would imply that  $\mathbb{Q}$  is cyclic. But if  $q \neq 0$ , then  $q/2 \notin \langle q \rangle$ , while  $\langle 0 \rangle = \{0\}$ . Thus  $\mathbb{Q}$  is not cyclic and therefore is not finitely generated.

□

**Exercise 2.4.15.** Exhibit a proper subgroup of  $\mathbb{Q}$  which is not cyclic.

*Solution.* Let

$$H = \left\{ \frac{m}{2^k} \mid m \in \mathbb{Z}, k \in \mathbb{N}_0 \right\}.$$

If  $m/2^k, n/2^\ell \in H$ , then

$$\frac{m}{2^k} - \frac{n}{2^\ell} = \frac{2^\ell m - 2^k n}{2^{k+\ell}} \in H,$$

so  $H \leq \mathbb{Q}$ . It is proper because  $1/3 \notin H$ .

Suppose  $H = \langle q \rangle$  for some  $q \in H$ . If  $q = 0$ , then  $\langle q \rangle \neq H$ . If  $q = m/2^k \neq 0$ , then

$$\frac{1}{2^{k+1}} \in H$$

cannot be an integer multiple of  $q$ , since

$$n \frac{m}{2^k} = \frac{1}{2^{k+1}}$$

would imply  $2nm = 1$ . Hence  $H$  is not cyclic.

□

**Exercise 2.5.3.** Show that the subgroup  $\langle s, r^2 \rangle$  of  $D_4$  is isomorphic to  $V_4$ .

*Solution.* Since  $r^4 = s^2 = 1$  and  $sr^2 = r^2s$ , we have

$$\langle s, r^2 \rangle = \{1, r^2, s, sr^2\}.$$

Define

$$\varphi: (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}) \rightarrow \langle s, r^2 \rangle, \quad \varphi([i]_2, [j]_2) = r^{2i}s^j.$$

Because  $r^2$  and  $s$  commute and both have order 2,  $\varphi$  is a homomorphism. Its four values are precisely

$$1, r^2, s, sr^2,$$

so it is bijective. Since  $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}) \cong V_4$ ,

$$\langle s, r^2 \rangle \cong V_4.$$

□

**Exercise 2.5.9.** Draw the lattices of subgroups of the following groups:

- (a)  $\mathbb{Z}/16\mathbb{Z}$ ,
- (b)  $\mathbb{Z}/24\mathbb{Z}$ ,
- (c)  $\mathbb{Z}/48\mathbb{Z}$ .

*Hint.* See Exercise 6 in Section 3.1.

*Solution.* For each divisor  $d$  of  $n$ , the cyclic group  $\mathbb{Z}/n\mathbb{Z}$  has a unique subgroup of order  $d$ , namely

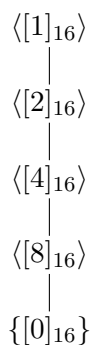
$$H_d = \langle [n/d]_n \rangle,$$

and

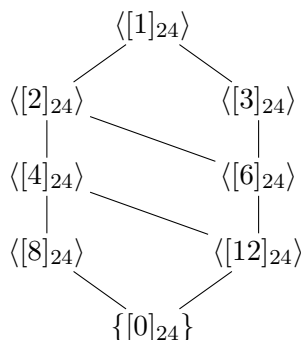
$$H_d \leq H_e \iff d \mid e.$$

Thus the subgroup lattices are the divisor lattices of 16, 24, and 48.

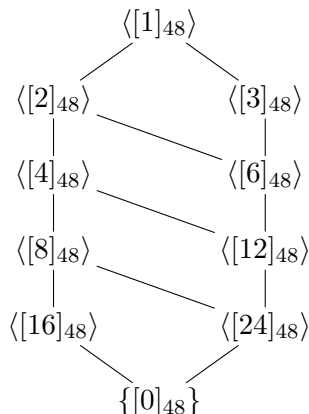
(a)



(b)



(c)



□

**Exercise 3.1.1.** Let  $G$  and  $H$  be groups. Let  $\varphi: G \rightarrow H$  be a homomorphism and let  $E$  be a subgroup of  $H$ . Prove that  $\varphi^{-1}(E) \leq G$ , i.e., the preimage or pullback of a subgroup under a homomorphism is a subgroup. If  $E \trianglelefteq H$ , prove that  $\varphi^{-1}(E) \trianglelefteq G$ . Deduce that  $\ker \varphi \trianglelefteq G$ .

*Solution.* Since  $\varphi(1_G) = 1_H \in E$ , the set  $\varphi^{-1}(E)$  is nonempty. If  $x, y \in \varphi^{-1}(E)$ , then

$$\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} \in E,$$

so  $xy^{-1} \in \varphi^{-1}(E)$ . By the Subgroup Criterion,

$$\varphi^{-1}(E) \leq G.$$

Now suppose  $E \trianglelefteq H$ . If  $g \in G$  and  $x \in \varphi^{-1}(E)$ , then

$$\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g)^{-1} \in E,$$

because  $E$  is normal in  $H$ . Hence

$$\varphi^{-1}(E) \trianglelefteq G.$$

Taking  $E = \{1_H\} \trianglelefteq H$  gives

$$\ker \varphi = \varphi^{-1}(\{1_H\}) \trianglelefteq G.$$

□

**Exercise 3.1.2.** Let  $\varphi: G \rightarrow H$  be a homomorphism of groups with kernel  $K$ , and let  $a, b \in \varphi(G)$ . Let  $X \in G/K$  be the fiber above  $a$  and let  $Y$  be the fiber above  $b$ , i.e.,

$$X = \varphi^{-1}(a), \quad Y = \varphi^{-1}(b).$$

Fix an element  $u \in X$ , so  $\varphi(u) = a$ . Prove that if  $XY = Z$  in the quotient group  $G/K$  and  $w$  is any member of  $Z$ , then there is some  $y \in Y$  such that  $uy = w$ .

*Hint.* Show that  $u^{-1}w \in Y$ .

*Solution.* By the description of fibers in §3.1,  $X = uK$ . Choose  $v \in Y$ , so  $Y = vK$ . Since

$$Z = XY = (uK)(vK) = uvK,$$

any  $w \in Z$  has the form

$$w = uvk$$

for some  $k \in K$ . Set

$$y = u^{-1}w = vk.$$

Since  $vk \in vK = Y$ , we have  $y \in Y$ , and by construction

$$uy = w.$$

□

**Exercise 3.1.7.** Define  $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}$  by

$$\pi((x, y)) = x + y.$$

Prove that  $\pi$  is a surjective homomorphism and describe the kernel and fibers of  $\pi$  geometrically.

*Solution.* For  $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ ,

$$\begin{aligned}\pi((x_1, y_1) + (x_2, y_2)) &= \pi((x_1 + x_2, y_1 + y_2)) \\ &= x_1 + x_2 + y_1 + y_2 \\ &= \pi((x_1, y_1)) + \pi((x_2, y_2)).\end{aligned}$$

Thus  $\pi$  is a homomorphism. It is surjective because for every  $c \in \mathbb{R}$ ,

$$\pi((c, 0)) = c.$$

The kernel is

$$\ker \pi = \{(x, y) \in \mathbb{R}^2 \mid x + y = 0\} = \{(t, -t) \mid t \in \mathbb{R}\},$$

the line through the origin with slope  $-1$ . More generally, the fiber above  $c \in \mathbb{R}$  is

$$\pi^{-1}(c) = \{(x, y) \in \mathbb{R}^2 \mid x + y = c\},$$

which is the line of slope  $-1$  with  $y$ -intercept  $c$ . Hence the fibers are parallel affine lines, each a coset of  $\ker \pi$ .

□